

## Design Goal:

The goal of this project is to design and fabricate two ground-mounted enrichment toy hangers for the Nashville Zoo that can:

- Support  $\geq 50$  lb suspended load
- Resist dynamic eland force
- Extend safely over the 8 ft fence by 1-2 feet for a 1.5 foot device.
- Prevent horn entanglement
- Allow staff to attach toys from outside the enclosure.
- Withstand outdoor weather exposure and repeated animal interaction.

Our priorities are in order of highest priority at the top and diminishing priority as it goes down.

## Priorities:

### 1. Effectiveness (High Priority)

- The toy hanger must safely support a suspended enrichment toy weighing at least 50 pounds for a minimum of 12 consecutive hours without structural failure or permanent deformation.
- Effectiveness is defined as the ability to:
  - Change the height of the toy for ease of changing (at least 4 feet above fence to 5 feet above the ground).
  - Extend the toy 1-2 feet beyond the fence (Toys can be up to 1.5 feet)
  - Place the toy's center about 5 feet off of the ground.
  - Carry 50 lbs.
- This requirement supports the primary stakeholders, the elands and animal care staff, by ensuring enrichment remains available and safe during use by the eland. Failure at any point could result in animal injury, staff risk, or loss of enrichment access, directly compromising animal welfare.

### 2. Structural Integrity (High Priority)

- The hanger must extend over the 8-foot enclosure fence while maintaining vertical alignment within 3 degrees of perpendicular to the ground under full load. To do this, the structure must:
  - Be inserted into the ground to prevent tipping.<sup>2</sup>
  - Resist bending moments caused by the eland activity and toy weight.<sup>3</sup>
  - Remain free of cracks, fractures, or weld separation.<sup>4</sup>

- Integrity is defined as the hanger maintaining structural stability under both static (50 pound hanging load) and dynamic loading conditions (pulling, lifting, and lateral forces from elands) without bending greater than 3 degrees of total tipping. The 50-pound requirement is based directly on the maximum enrichment toy weight specified by the Nashville Zoo. Because elands such as Loki and Olive can weigh 1200+ pounds and generate additional force when interacting with enrichment devices, the hanger must tolerate forces exceeding the toy's static weight<sup>1</sup>.
- This criterion reflects stakeholder value by ensuring enclosure safety and long-term mechanical reliability. Zoo staff require a structure that will not lean, collapse, or damage the fence, and the animals require a stable enrichment source that does not unpredictably shift during use.

### **3. Ability to Change Toys - Outside (High Priority)**

- The toy hanger must allow zoo staff to attach and remove enrichment toys entirely from outside the enclosure within 3 minutes without specialized tools. To do this, the attachment mechanism must:
  - Prevent accidental detachment by using a carabiner or eye bolt.
  - Be available outside the fence.
  - Maintain safe clearance above the fence while toys are attached.
  - Take less than 3 minutes to attach.
- This requirement exists because animal care staff cannot safely enter the enclosure during routine enrichment changes. Toys are often suspended with chains, which increase entanglement risk if improperly positioned.
- This requirement directly reflects stakeholder priorities of staff safety and operational efficiency. A system that requires entering the enclosure would increase injury risk and slow daily care routines.

### **4. Durability (High Priority)**

- The toy hanger must remain structurally functional for a minimum of 10 years of outdoor exposure without more than 10% material degradation or reduction in load-bearing capacity. To meet this requirement, the hanger must have:
  - No corrosion that compromises the structural integrity.
  - No hardware failure due to repeated stress.

- The device will remain outdoors year-round and be exposed to rain, humidity, and temperature fluctuations.<sup>5</sup> Additionally, repeated animal interaction creates cyclic loading that can fatigue materials over time.<sup>5</sup>
- This requirement supports zoo administration and maintenance stakeholders by reducing replacement frequency and long-term costs while ensuring continuous enrichment access for the animals.

## **5. Low Price (High Priority)**

- The device should minimize material and fabrication costs while meeting all structural and safety requirements.
- The design should minimize unnecessary complexity or materials that would significantly increase cost to meet the \$250 budget.
- On average, the cost of a similar device is \$6208.33 when choosing  $n = 3$  samples<sup>6</sup>
- This requirement supports broader accessibility and adoption, which supports stakeholder concerns because the stakeholders need to be able to afford the device.

## **6. Ease of Maintenance (Low Priority)**

- The hanger should allow visual safety inspection within 5 minutes and require no more than two standard tools for hardware replacement. The design should<sup>7</sup>:
  - Minimize moving parts.
  - Avoid debris trapping crevices.
  - Use standardized hardware sizes when possible.
- This aligns with stakeholder needs for operational efficiency and long-term usability.

## **7. Aesthetics (Low Priority)**

- The hanger should maintain a clean, professional appearance with smooth welds and no exposed sharp edges.
- Because the installation is located behind the scenes and not on public exhibit, aesthetics are secondary to safety and functionality. This reflects stakeholder priorities by ensuring resources are directed toward structural performance rather than cosmetic enhancements.

## Stakeholder Validation:

Zoo Personnel 1:



## Citations:

1. Kyle Banton-Jones, No Date, "Hanging Food Enrichment Barrel -", No Publication, <https://wildenrichment.com/bears/hanging-food-enrichment-barrel/>
2. Electrical Contractor 23, 11-15-2023, "Building a Solid Foundation: A strong base in Code application for grounding electrodes", Default, <https://www.ecmag.com/magazine/articles/article-detail/building-a-solid-foundation-a-strong-base-in-code-application-for-grounding-electrodes>
3. K.J. Pascoe, 1978, "An Introduction to the Properties of Engineering Materials", SpringerLink, <https://link.springer.com/book/10.1007/978-94-011-7068-0>
4. FHWA, 1999 "GeoTechnical Engineering Circular No. 4", US Department of Transportation, <https://www.fhwa.dot.gov/engineering/geotech/pubs/if99015.pdf>
5. Lapis 25, 9-11-2025, "Durable Patio Furniture for Rain: Best Materials That Last – Lapis Patios", No Publication, <https://lapispatios.com/articles/durable-patio-furniture-for-rain/>
6. See Part B
7. Choi Design 24, 11-7-2024, "The Power of Minimalism: Why Less is More in Modern Product Design – Choi Design", <https://choidesign.com/the-power-of-minimalism-why-less-is-more-in-modern-product-design/>